IOT ENABLED 3-PHASE POWER QUALITY & ENERGY METER

TUE-17

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CHALLENGE PROBLEM

CONVENTIONAL METERS:

1. Show all values in real-time, no logging for future reference
2. Lack critical parameters like power factor, THD
3. Have no provision for data visualization
4. Lack conditional assessment of various loads

SOLUTION

DATA LOGGING

Data logging functionality to store data on server

CRITICAL PARAMETERS

Availability of various critical parameters like THD, PF, VAR

DATA VISUALIZATION

On-screen data visualization functionality

IOT FUNCTIONALITY

IOT dashboard for analytics

VALUE PROPOSITION

1. Enables logging and retrieval of past data for better analysis and decision-making
2. Ensures remote monitoring and control, allowing users to access data from anywhere
3. Facilitates conditional assessment of various loads to optimize power usage and prevent failures

TARGET SPECIFICATIONS

1. Raw current and voltage logging for 3-phase supply
2. Availability of other critical power quality parameters
3. Data visualization dashboard
4. Harmonics measurement in current and voltage waveforms
5. Remote control of loads (turning ON/OFF)

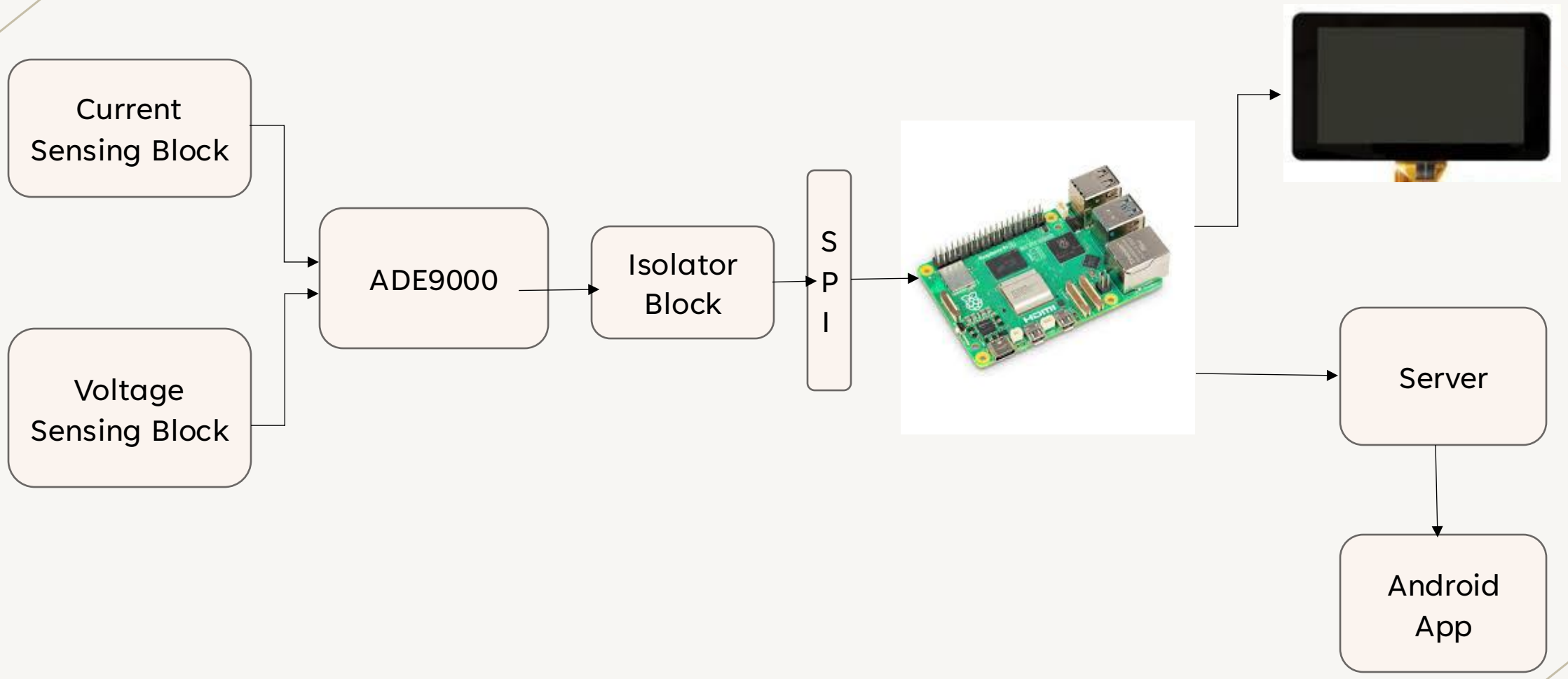
SOLUTION APPROACH

1. ADE9000 can be used for sampling current and voltage waveforms as well as for other critical power quality parameters
2. Raspberry pi can be used for data processing, cloud connectivity, real-time visualization
3. Server integration for data logging
4. Display for data visualization with multiple tabs
5. Relay will be used for remote control of loads

PRINCIPLE OF OPERATION

1. ADE9000 samples voltage and current waveforms via connected sensors (CTs/PTs).
2. Built-in ADCs convert analog signals into digital data for processing
3. ADE9000 processes sampled data and computes power quality parameters
4. Raspberry Pi communicates with ADE9000 via SPI to fetch processed data
5. Raspberry Pi forwards data to a cloud server via Wi-Fi/Ethernet for remote access
6. Data is stored and analyzed for historical trends, alerts, and visualization

BLOCK DIAGRAM



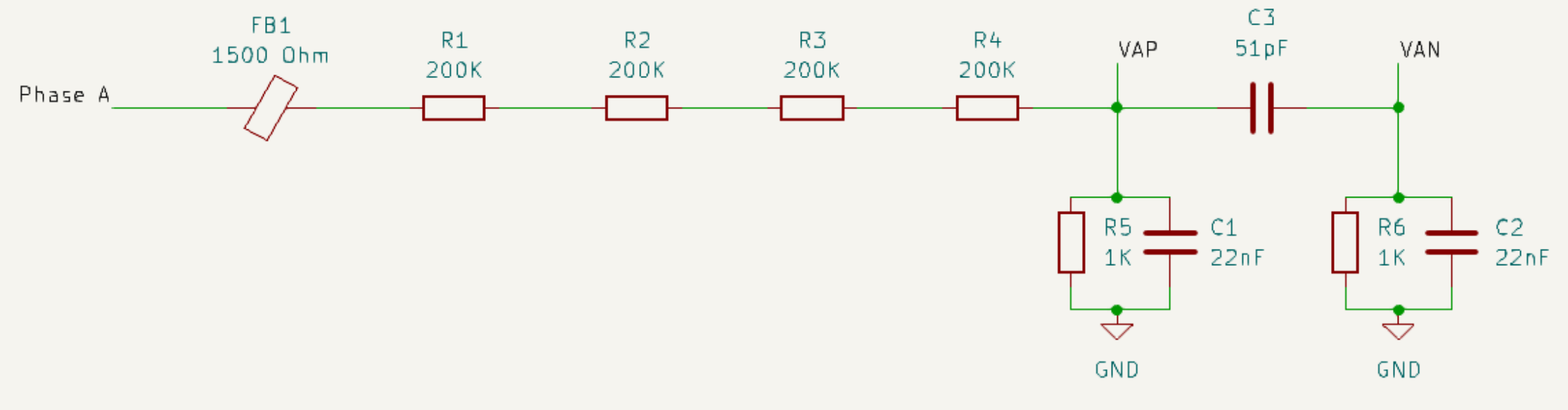
BILL OF MATERIALS (BOM)

Components	Description
Raspberry Pi 5	Main controller
Display 7' Touch	Display
ADE9000	Power Monitoring
ADUM4151BRIZ	Isolator
ADUM6404ARIZ	Isolator
SN74LV4T125PWR	Level Shifter
24.576 MHz Crystal	Crystal Oscillator
Relays	Turning load on/off
Current Transformer	Current Sensing
Resistors	For various blocks
Capacitors	For various blocks
ADP122	LDO

Components	Description
Transistor (BC 547)	For driving relay
Ferrite Bead	Used in Voltage Block
Micro USB Connector	For Powering Board
Screw Terminals	Connectors
Female Headers	Connectors
Diodes	For current block
Legrand Casing Box	For covering the setup

CIRCUIT DIAGRAMS

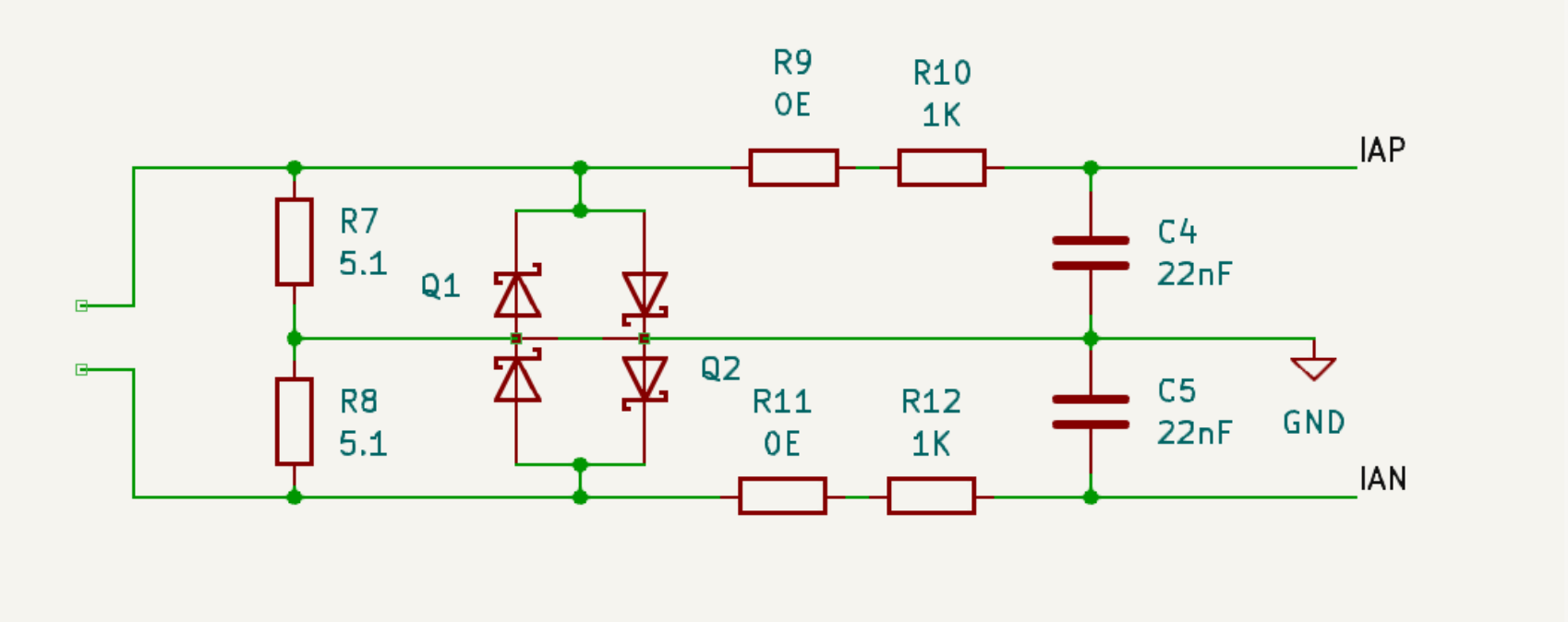
Voltage Sensing Block



There will be 3 such blocks for each of 3 phase

CIRCUIT DIAGRAMS

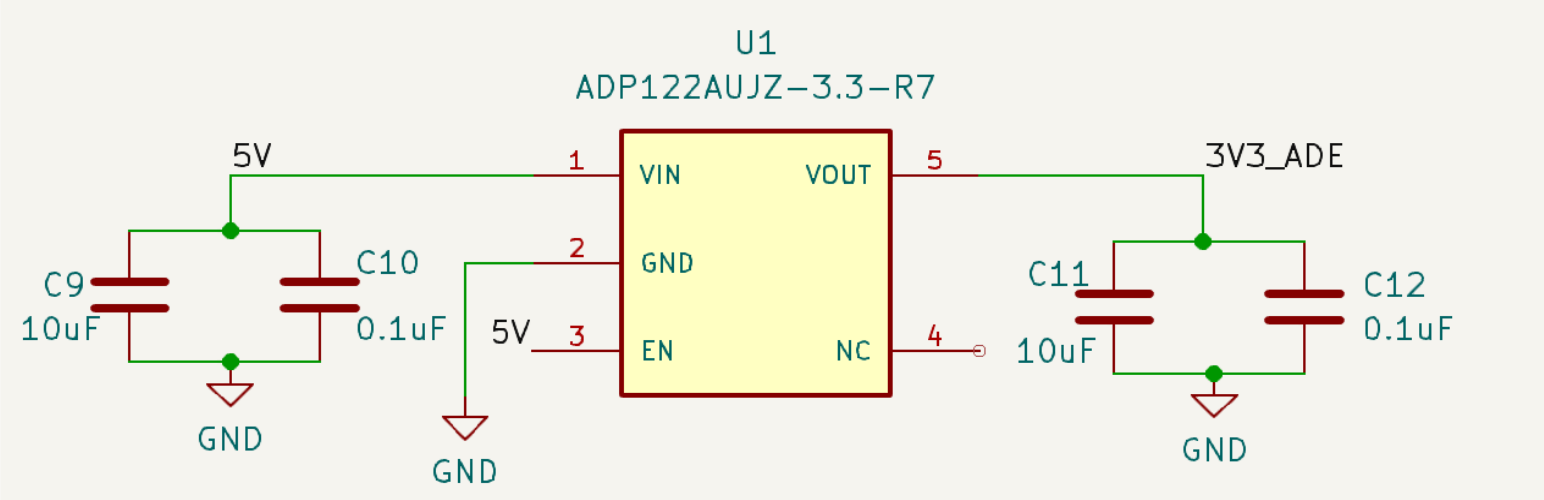
Current Sensing Block



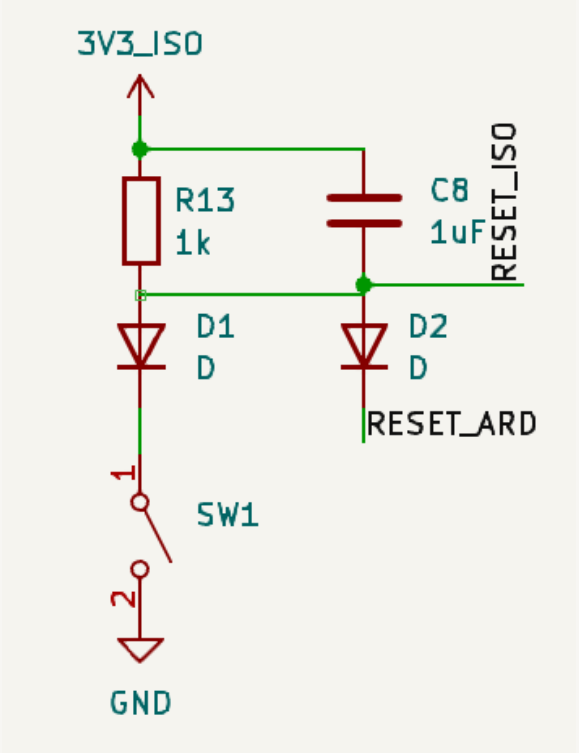
There will be 4 such blocks for each of 3 phase and neutral

CIRCUIT DIAGRAMS

Voltage Regulator

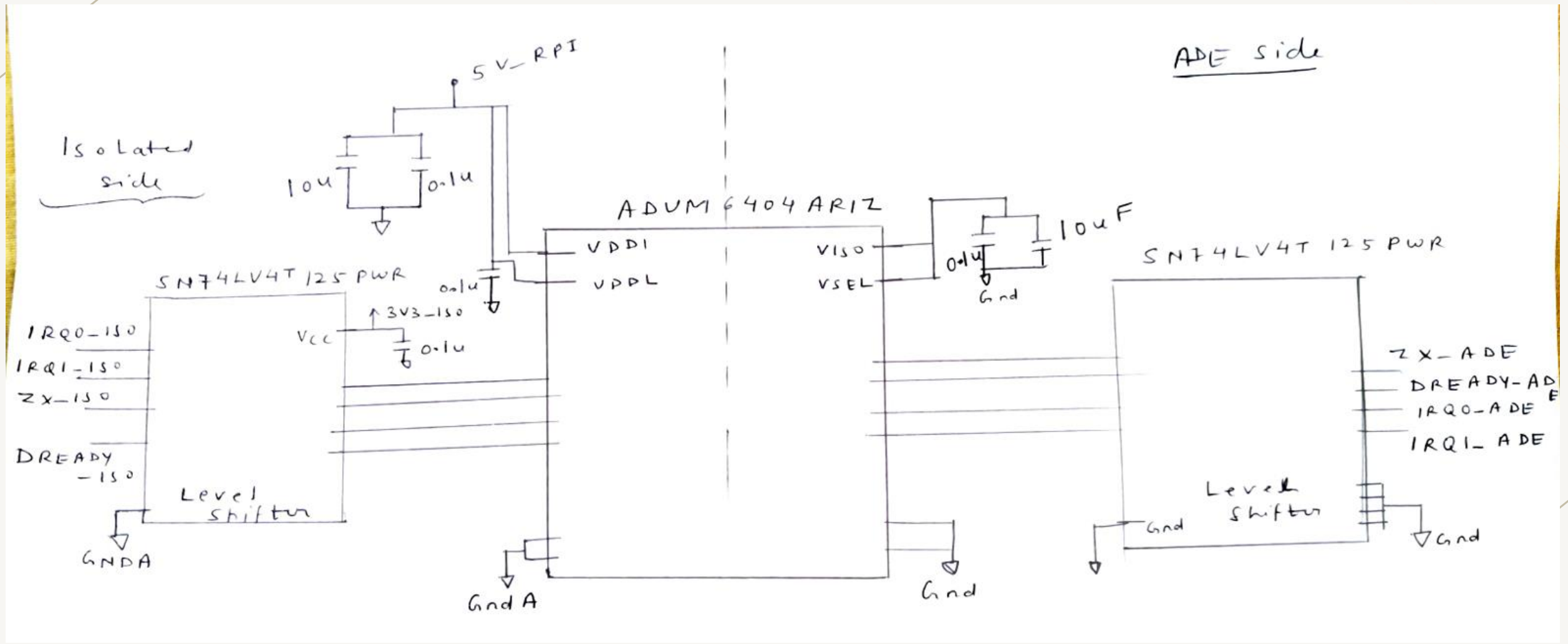


RESET Block



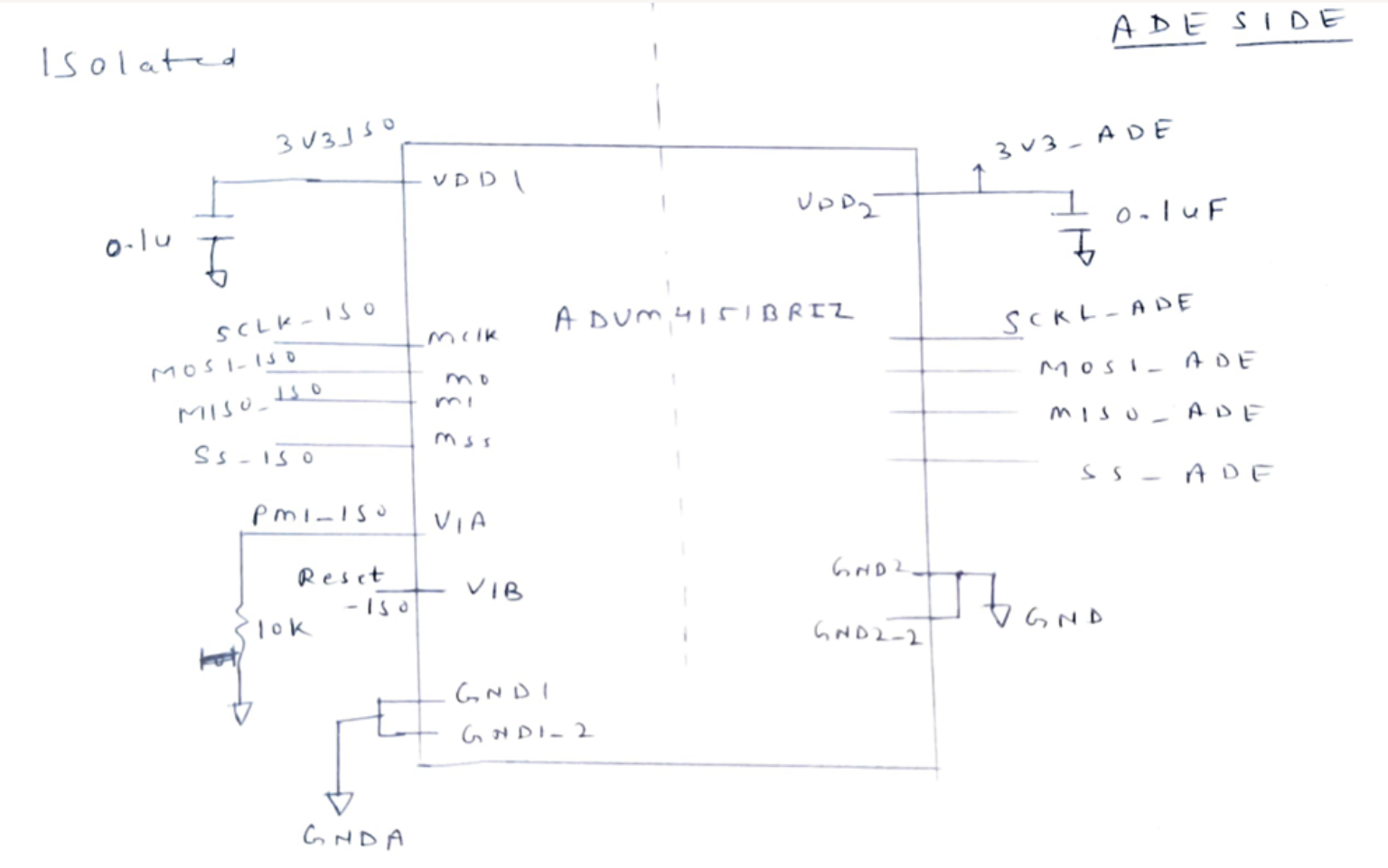
CIRCUIT DIAGRAMS

Isolator Block

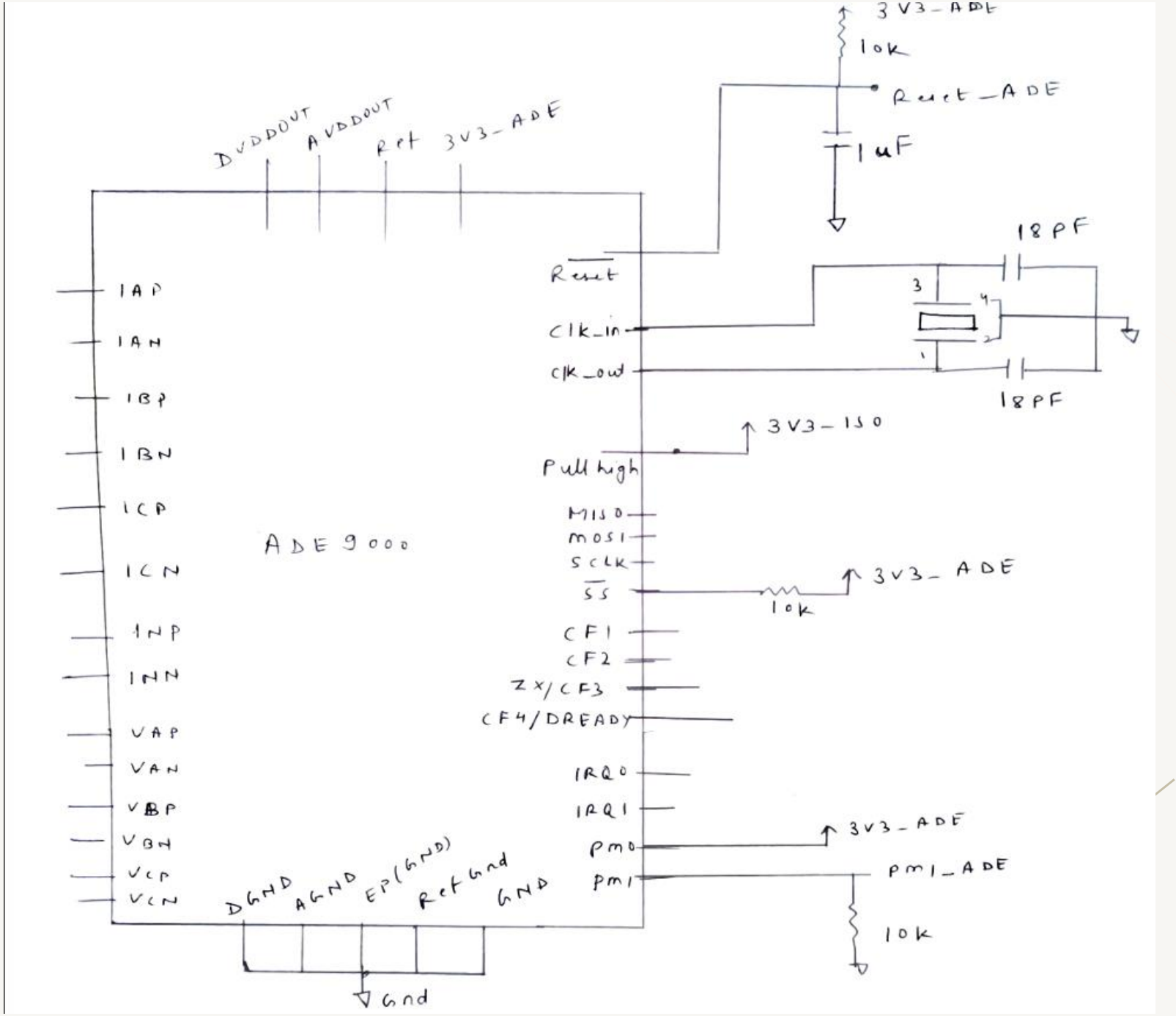


CIRCUIT DIAGRAMS

Isolator Block



ADE9000


3V3_ADE
10k
Reset_ADE
1uF
18PF
DVDDOUT
AVDDOUT
Ref 3V3_ADE
IAP
IAN
IBP
IBN
ICP
ICN
INP
INN
VAP
VAN
VBP
VBN
VCP
VCN
ADE9000
Reset
Clk_in
Clk_out
18PF
3V3_ISO
Pull high
MISO
MOSI
SCLK
SS
10k
3V3_ADE
CF1
CF2
ZX/CF3
CF4/DREADY
IRQ0
IRQ1
PM0
PM1
3V3_ADE
PM1_ADE
10k
DGND
AGND
EP(GND)
Ref Gnd
GND
Gnd

WORK DISTRIBUTION

SAURABH KUMAR

Server integration and Application development

SARTHAK NIRANJAN

Application development and UI development for Raspberry pi

ARYAN TAMBOLI

Testing ADE9000 IC using evaluation board, Circuit schematics (CAD) and PCB layout

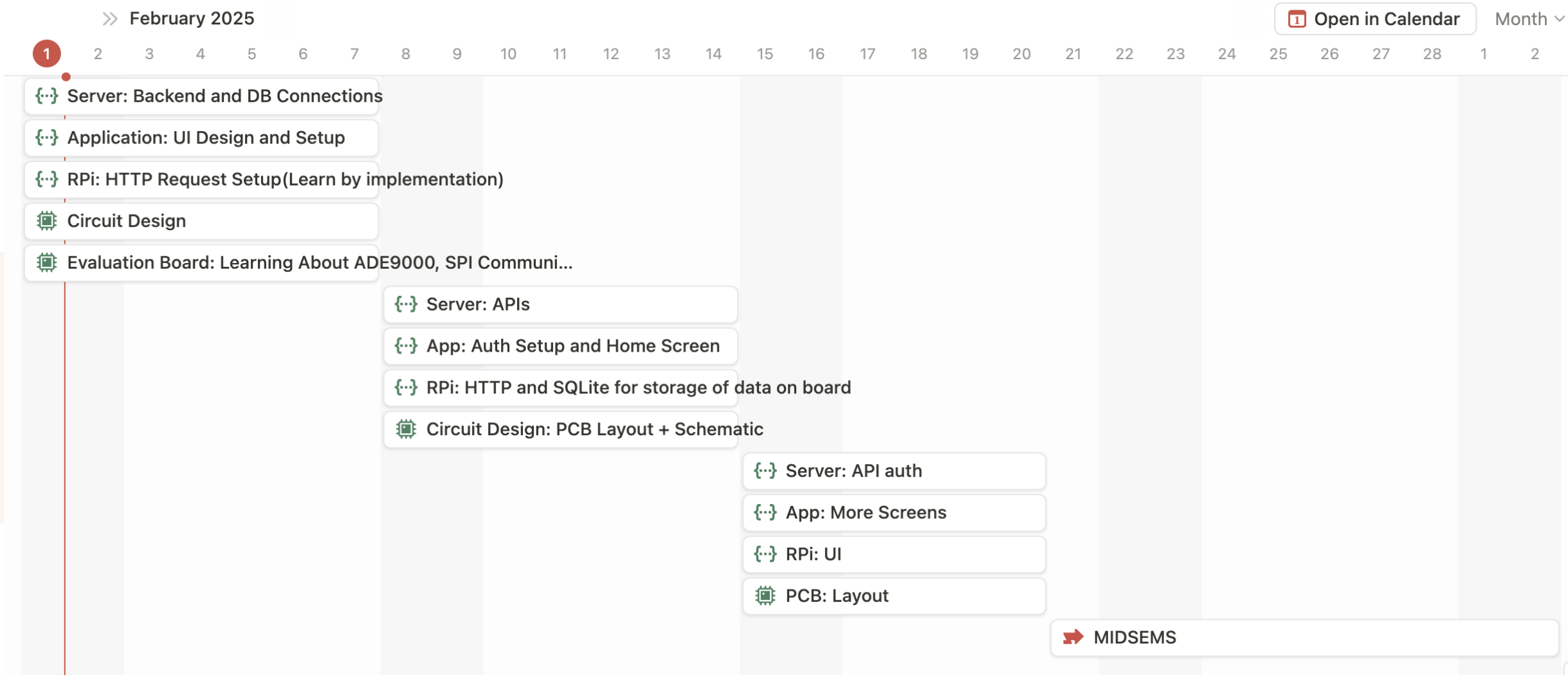
CHERISH JAIN

Integrating Raspberry pi with ADE9000 IC

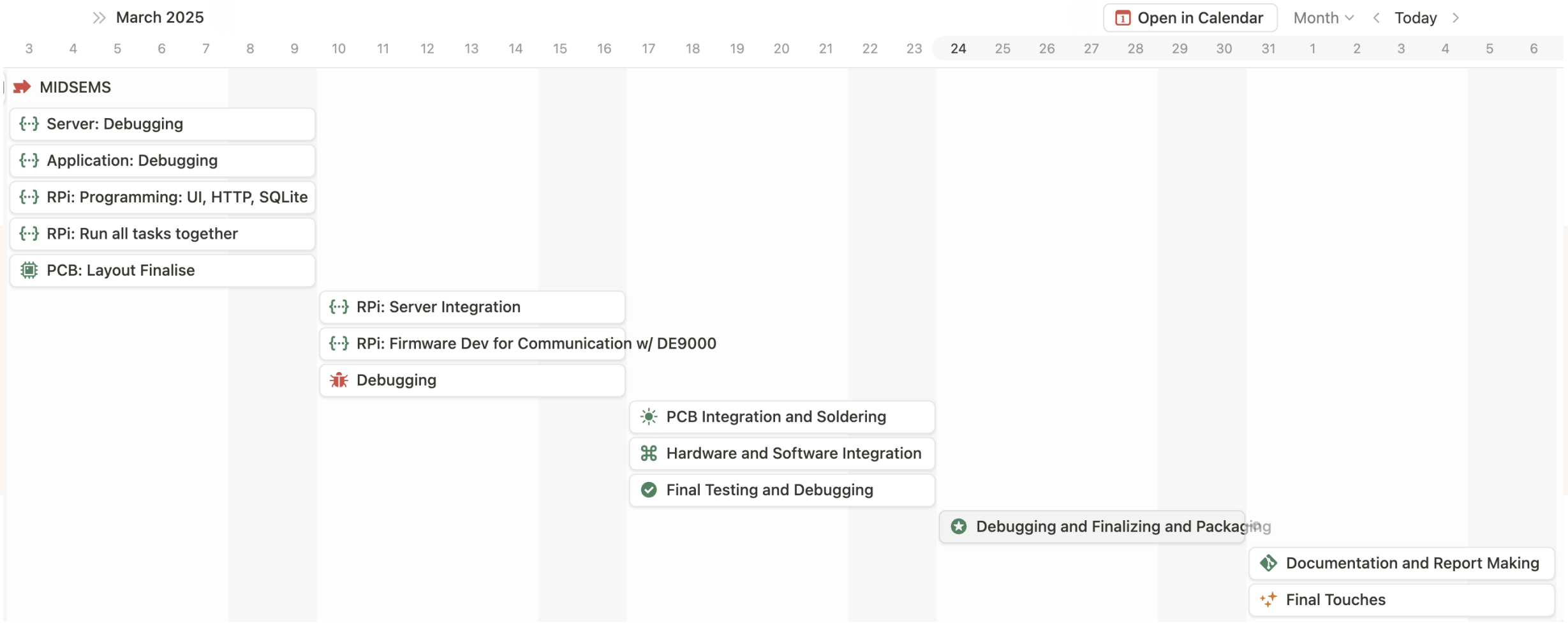
AYUSH GAUR

Analog front end, ADE9000 testing and PCB layout

WEEK-WISE GOALS



WEEK-WISE GOALS



KEY RISKS AND MITIGATION STRATEGIES

1. ***RISK***: Sensor noise and improper calibration can cause inaccurate readings from ADE9000

   MITIGATION: Implementation of proper shielding and periodic calibration

2. ***RISK***: Data corruption or loss due to unstable SPI communication

   MITIGATION: Implementation of CRC checks and optimized SPI clock speed

3. ***RISK:*** Network failure or server downtime may lead to missing data logs

   MITIGATION: Local caching on Raspberry Pi and automatic re-upload

4. ***RISK:*** Reliance on single-phase power draw may disable monitoring of the remaining two phases if that phase fails.

   MITIGATION: Integrate a standby battery to ensure continuous board operation during power outages.

DELIVERABLES

1. Fully functional Power meter capable of showing on-screen data plots.
2. Remote load control
3. Optimized software stack
4. IOT Integration
5. Build Android application for tracking data
6. Resolved bugs (from previous design)

THANK YOU